# Semantic Web technologies: a memento

| ACRONYM | FULL NAME | WHAT IT IS | GOOD FOR |
|---|---|---|---|
| 1 · Representing data<br>**RDF** | **Resource Description Framework** | The basic data model of the Semantic Web. It represents everything as *triples* (subject – predicate – object), such as :Ana :teaches :KRR, with resources identified by IRIs. Many triples together form a graph. | Describing and linking data from different sources in one uniform format, which is the foundation for everything below. |
| 2 · Modeling meaning<br>**RDFS** | **RDF Schema** | A small vocabulary on top of RDF for basic structure: classes, subclasses, properties, subproperties, and the domain and range of properties. | Simple taxonomies and lightweight inference, such as “every Professor is a Person.” |
| 2 · Modeling meaning<br>**OWL** | **Web Ontology Language** | A much more expressive ontology language, grounded in Description Logics. It adds cardinality, equivalence, disjointness, inverse and transitive properties, restrictions, and more. | Building rich ontologies and letting a reasoner infer new facts and detect inconsistencies. It assumes an *open world*. |
| 3 · Reasoning with rules<br>**SWRL** | **Semantic Web Rule Language** | A language for Horn-like rules (IF … THEN …) over OWL ontologies, e.g. Parent(?x) ∧ Brother(?y, ?x) → Uncle(?y). A W3C Member Submission that never became a full standard, but supported in Protégé and by reasoners like Pellet. | Expressing inferences that OWL alone cannot capture. |

**Blue: Semantic Web stack**

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| 4 · Querying<br>**SPARQL** | **SPARQL Protocol and RDF Query Language** | The standard query language for RDF graphs, roughly “SQL for graphs.” It works by matching triple patterns. | Retrieving, filtering, aggregating and updating data in RDF stores and public endpoints such as Wikidata or DBpedia. |
| 5 · Validating data<br>**SHACL** | **Shapes Constraint Language** | A language for defining *shapes*, constraints RDF data must satisfy, such as “every Person must have exactly one name of type string.” It assumes a *closed world*. | Checking data quality and completeness in RDF graphs and producing validation reports. |
| 6 · Exchanging data<br>**JSON-LD** | **JSON for Linked Data** | A way to write RDF as ordinary JSON. An @context maps JSON keys to IRIs, so the same file is both regular JSON and an RDF graph. | Publishing linked data on the web and in APIs (e.g. schema.org markup), without developers having to learn RDF. |
| 6 · Exchanging data<br>**JSON Schema** | **JSON Schema** | A vocabulary for describing the expected structure of plain JSON: required fields, data types, value ranges and patterns. Not a Semantic Web technology. | Validating JSON in APIs and configuration files. It checks *syntax and structure*, not *meaning*. |

**Blue: Semantic Web stack** · **Orange: JSON world**

**Common confusions**

**RDFS vs. OWL:** both describe vocabularies, but OWL is far more expressive. RDFS gives hierarchies, while OWL gives logic

**OWL vs. SHACL:** OWL *infers* facts in an open world, where missing information means unknown. SHACL *checks* data in a closed world, where missing information means a violation

**SHACL vs. JSON Schema:** both validate data. SHACL validates RDF graphs, which carry meaning. JSON Schema validates JSON trees, which only have structure

**RDF vs. JSON-LD:** RDF is the data model, and JSON-LD is one way of writing it down. Turtle and RDF/XML are other ways